SILVER SULFADIAZINE

Burn wounds

Sulfa allergy; avoid in neonates

CHEMICAL NAME Silver 4-amino-N-(pyrimidin-2-yl)benzenesulfonamide
CLASS Topical sulfonamide antibacterial (burns)
TRADE NAMES Flamazine · Silvadene · SSD cream · Dermazin

THERAPEUTIC USES / INDICATIONS

- Prevention and treatment of infection in partial- and full-thickness burns.
- Leg ulcers, pressure sores, infected wounds (short-term).

PHARMACODYNAMICS

- Silver ions damage **bacterial cell membranes and DNA**; broad spectrum including *Pseudomonas*, *Staph. aureus*, Candida.

ADMINISTRATION

- **Topical:** apply 3–5 mm layer with sterile gloved hand/spatula to cleaned burn, cover with dressing; change daily (or more if exudate).
- Remove old cream residue at each dressing change.
- Avoid large areas for prolonged periods (systemic absorption).

SIDE & ADVERSE EFFECTS

SIDE EFFECTS (COMMON)

- Burning, itching, pain on application.
- Grey/blue skin discoloration (argyria).
- Pseudo-eschar formation (delays assessment).

ADVERSE EFFECTS (SERIOUS)

- Transient leucopenia (first days).
- Systemic sulfonamide reactions: rash, SJS, haemolysis (G6PD), crystalluria.
- Delayed wound healing (cytotoxic to keratinocytes).

DOSAGES

- 1% cream applied once daily (up to twice daily) to burns.
- Use until healing or grafting.

PHARMACOKINETICS

- Silver acts locally; sulfadiazine can be absorbed from large burns (up to 10%) → systemic effects.
- Absorbed sulfonamide excreted in urine.

CONTRAINDICATIONS & CAUTIONS

- Sulfonamide hypersensitivity.
- Pregnancy near term, premature infants, neonates <2 months (kernicterus).
- G6PD deficiency (caution).
- Facial burns (prefer alternatives).

KEY INTERACTIONS

- **Enzymatic debriding agents:** inactivated.
- **Large burns:** may potentiate oral hypoglycaemics, phenytoin (systemic absorption).

NURSING MANAGEMENT

- Aseptic dressing technique; assess burn depth, exudate, infection signs.
- FBC (leucocytes) in extensive burns.
- Pain management before dressing changes.
- Check sulfa allergy and age.

PATIENT EDUCATION

- Dressings are changed daily; report increasing pain, fever, rash.
- Skin may turn grey temporarily.

KEY POINTS

- Topical silver + sulfonamide for burns.
- Avoid in sulfa allergy, neonates, late pregnancy.
- 3–5 mm layer, daily change.
- Transient leucopenia.